



Vivekanand Education Society's Institute Of Technology
Department Of Information Technology
DSA mini Project
A.Y. 2025-26

Title: Airport Check-in Counter Simulation
Sustainability Goal : Reduced Inequalities

Domain:

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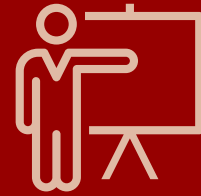
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Introduction to Project



Introduction to Airport Check-in Simulation-

Airports handle thousands of passengers every day, and one of the most critical operations is the **check-in process**. This involves passengers arriving, waiting in queues, verifying their tickets, dropping off baggage, and finally receiving their boarding passes.

Why Simulation?

A simulation allows us to **model real-world systems** like an airport check-in counter in a controlled environment. By simulating passenger arrivals, queues, and processing, we can study how efficiently the system works and identify bottlenecks.



Problem Statement

Airports experience a high volume of passengers daily, making the check-in process one of the most critical and time-consuming tasks.

Passengers must queue at different counters, verify their identity, submit baggage, and collect their boarding passes. However, delays often occur due to **long queues, uneven counter allocation, and priority passengers** needing faster service.

In order to optimize this process, there is a need to design a **computer-based simulation of airport check-in** using concepts of **Data Structures and Algorithms (DSA)**. This simulation will model passengers arriving, joining queues, being served at counters, and leaving the system.

Objectives of the project



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To simulate the airport check-in process

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To implement core Data Structures



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To analyze system efficiency

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To demonstrate real-world applicability of DSA



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To provide a flexible simulation

DATA STRUCTURES



- Priority Queue
- Singly Linked List
- Structures (struct)
- Global Pointers (front and rear)
- Counter Variable (nextId)



Requirements of the system (Hardware, software)

Software



Programming Language:C

IDE:IntelliJ IDEA

Compiler / Runtime–program wiz
online C compiler

Version Control :Git / GitHub

Simulation Output:Console-based
for queue and passenger
processing

o

Hardware



Processor: Intel i3

RAM: Minimum 4 GB

Storage: At least 1 GB free
space

Display: Standard monitor

Input Devices: Keyboard and
mouse

Operating System Support:
Windows / macOS



FRONT END

Airport Check-in Simulator

Refresh

Process Tick

Start Auto Mode

 Search passenger...

Counters

Counter 1

No passengers waiting

Counter 2

No passengers waiting

Counter 3

No passengers waiting

Analytics

Processed: 0

Waiting: 0

Avg Queue: 0.00

Counters: 3

Est. Wait: 0.0s

Recently Processed

No passengers processed yet...

Live Activity

No activity yet...

Add Passenger

Name

1

Add Passenger

Chatbot

Ask about queues, counters, stats

Send



CODE

```
const response = await axios.post(
  "https://generativelanguage.googleapis.com/v1beta/models/gemini-2.0-flash:generateContent",
  {
    contents: [
      {
        role: "user",
        parts: [{ text: `${context}\nUser: ${message}` }],
      },
    ],
  },
  {
    headers: { "Content-Type": "application/json" },
    params: { key: process.env.GEMINI_API_KEY },
  }
);
```

External Google API called for chatbot

```
class Queue {
  constructor() {
    this.items = [];
  }
  enqueue(item) {
    this.items.push(item);
  }
  dequeue() {
    return this.items.shift();
  }
  size() {
    return this.items.length;
  }
  isEmpty() {
    return this.items.length === 0;
  }
  getAll() {
    return this.items;
  }
}
```



CODE

```
export class AirportSimulator {
  constructor(numCounters = 3) {
    this.counters = Array.from({ length: numCounters }, () => new Queue());
    this.totalProcessed = 0;
    this.waitingTimes = [];
    this.history = [];
    this.logs = [];
  }

  assignPassenger(passenger) {
    const index = this.counters.reduce(
      (minIdx, q, i, arr) => (q.size() < arr[minIdx].size() ? i : minIdx),
      0
    );
    this.counters[index].enqueue(passenger);
    this.logs.push(` ✈ Passenger ${passenger.name} assigned to Counter ${index + 1}`);
    return index + 1;
  }

  process() {
    this.counters.forEach((queue, i) => {
      if (queue.size() > 0) {
        const p = queue.dequeue();
        this.totalProcessed++;
        this.logs.push(` ✅ Processed ${p.name} at Counter ${i + 1}`);
        this.history.push({
          name: p.name,
          flight: p.flight,
          baggageCount: p.baggageCount,
          counter: i + 1,
          processedAt: new Date(),
        });
      }
    });
  }
}
```

```
getStatus() {
  return {
    totalProcessed: this.totalProcessed,
    counters: this.counters.map((q, i) => ({
      id: i + 1,
      queueLength: q.size(),
      passengers: q.getAll() || [],
    })),
    logs: this.logs.slice(-10).reverse(),
  };
}

getAverageQueueLength() {
  const total = this.counters.reduce((a, q) => a + q.size(), 0);
  return (total / this.counters.length).toFixed(2);
}

export const airportSim = new AirportSimulator(3);
```



OUTPUT

Add Passenger

Shravani Mankar

1

Add Passenger

Add Passenger

Passenger assigned to counter

Counters

Counter 1

Shravani
Mankar

A1261 • 1

Counter 2

No passengers waiting

Counter 3

No passengers waiting

Analytics

Processed: 0

Waiting: 1

Avg Queue: 0.33

Counters: 3

Est. Wait: 3.3s

Live Activity

Passenger Shravani Mankar assigned to Counter 1



OUTPUT

Counters

Counter 1

Soham Mankar

SQ742 • 1

Counter 2

Shravani Mankar

EK243 • 1

Counter 3

No passengers waiting

Analytics

Processed: 0

Waiting: 2

Avg Queue: 0.67

Counters: 3

Est. Wait: 6.7s

Recently Processed

No passengers processed yet...

Live Activity

Passenger Shravani Mankar assigned to Counter 2

Passenger Soham Mankar assigned to Counter 1

Counters

Counter 1

Shravani Mankar

AI261 • 1

test abcd

QR252 • 1

Counter 2

Test user 123

EK316 • 5

Counter 3

dsa project test

AI346 • 2

Analytics

Processed: 0

Waiting: 4

Avg Queue: 1.33

Counters: 3

Est. Wait: 13.3s

Recently Processed

No passengers processed yet...

Live Activity

Passenger test abcd assigned to Counter 1

Passenger dsa project test assigned to Counter 3

Passenger Test user 123 assigned to Counter 2

Passenger Shravani Mankar assigned to Counter 1



OUTPUT

Process Tick

Counters		
Counter 1 👤 test abcd OK252 • 1	Counter 2 No passengers waiting	Counter 3 No passengers waiting
Analytics		
Processed: 3	Waiting: 1	Avg Queue: 0.33
Counters: 3	Est. Wait: 3.3s	
Recently Processed		
✔️ dia project test (A0340) – Counter 3		
✔️ Test user 123 (EX316) – Counter 2		
✔️ Shrawani Mankar (A0261) – Counter 1		
Live Activity		
✔️ Processed dia project test at Counter 3		
✔️ Processed Test user 123 at Counter 2		
✔️ Processed Shrawani Mankar at Counter 1		
🚨 Passenger test abcd assigned to Counter 1		
🚨 Passenger dia project assigned to Counter 3		

Start Auto Mode

Counters		
Counter 1 No passengers waiting	Counter 2 No passengers waiting	Counter 3 No passengers waiting
Analytics		
Processed: 4	Waiting: 0	Avg Queue: 0.00
Counters: 3	Est. Wait: 0.0s	
Recently Processed		
✔️ test abcd (GR252) – Counter 1		
✔️ dia project test (A0340) – Counter 3		
✔️ Test user 123 (EX316) – Counter 2		
✔️ Shrawani Mankar (A0261) – Counter 1		
Live Activity		
✔️ Processed test abcd at Counter 1		
✔️ Processed dia project test at Counter 3		
✔️ Processed Test user 123 at Counter 2		
✔️ Processed Shrawani Mankar at Counter 1		
🚨 Passenger test abcd assigned to Counter 1		



ADDITIONAL FUNCTIONALITIES

Airport Check-in Simulator

Shravani

Counters

Counter 1

No passengers waiting

Counter 2

Shravani Mankar

LH184 • 1

Counter 3

No passengers waiting

Chatbot

How many passengers are

How many passengers are waiting? There are currently 4 passengers waiting.

Ask about queues, counters, stats

Send



Conclusion

The Airport Check-in Simulation demonstrates how queues can effectively manage passenger flow in a real-world scenario. Using C language and linked list-based queue implementation, the project shows how passengers arrive, wait, and are served in a fair and systematic manner. This project highlights the practical use of data structures in solving everyday problems, while keeping the model simple and extendable for future improvements.



References

[1] A. S. Tanenbaum and D. J. Wetherall, Computer Networks, 5th ed. Upper Saddle River, NJ, USA: Prentice Hall, 2011.

[2] E. Horowitz, S. Sahni, and S. Anderson-Freed, Fundamentals of Data Structures in C, 2nd ed. Silicon Press, 2008.

